AGILE DEVELOPMENT AND DEVOPS LAB FAT

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Date : 15 / 04 / 2026

SET 4

Digital Wallet System

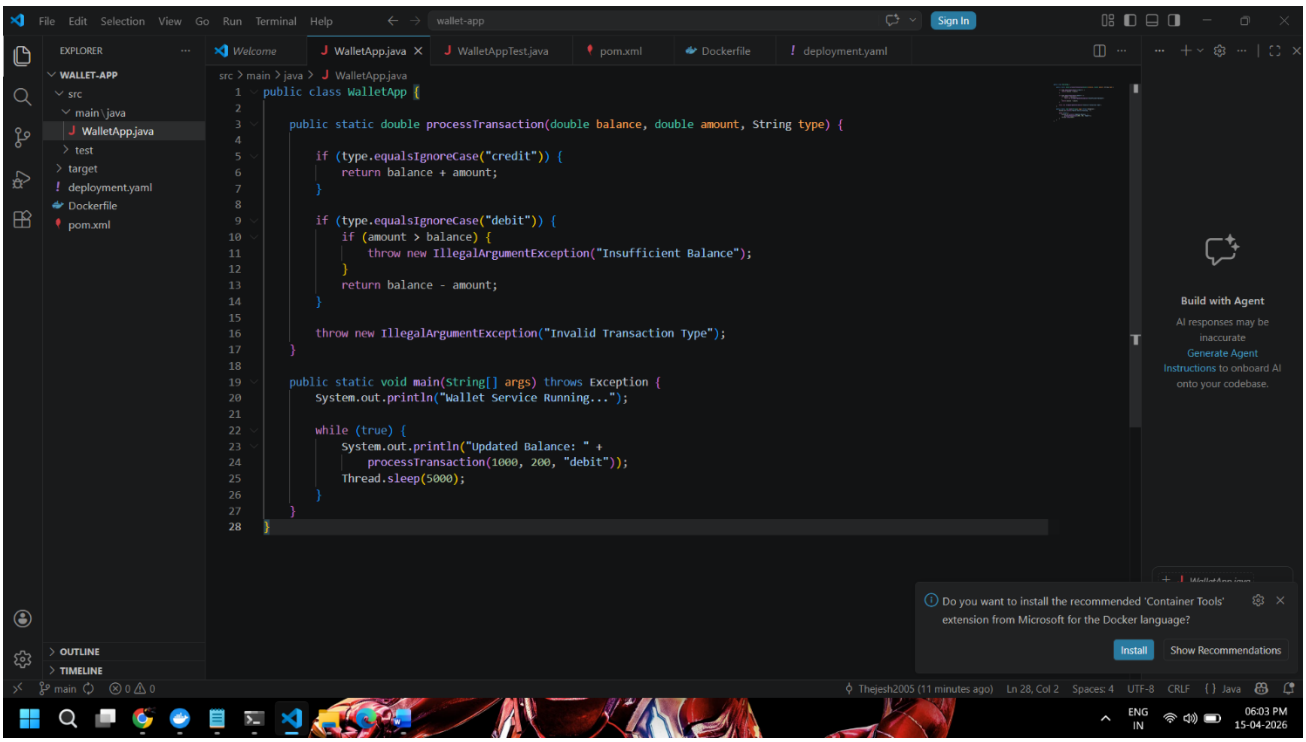
Tasks :

1. Develop Wallet Transaction Logic

   WalletApp.java

```
public class WalletApp {

    public static double processTransaction(double balance, double amount, String type) {

        if (type.equalsIgnoreCase("credit")) {
            return balance + amount;
        }

        if (type.equalsIgnoreCase("debit")) {
            if (amount > balance) {
                throw new IllegalArgumentException("Insufficient Balance");
            }
            return balance - amount;
        }

        throw new IllegalArgumentException("Invalid Transaction Type");
    }

    public static void main(String[] args) throws Exception {
        System.out.println("Wallet Service Running...");

        while (true) {
            System.out.println("Updated Balance: " +
                processTransaction(1000, 200, "debit"));
```

```
            Thread.sleep(5000);
        }
    }
}
```

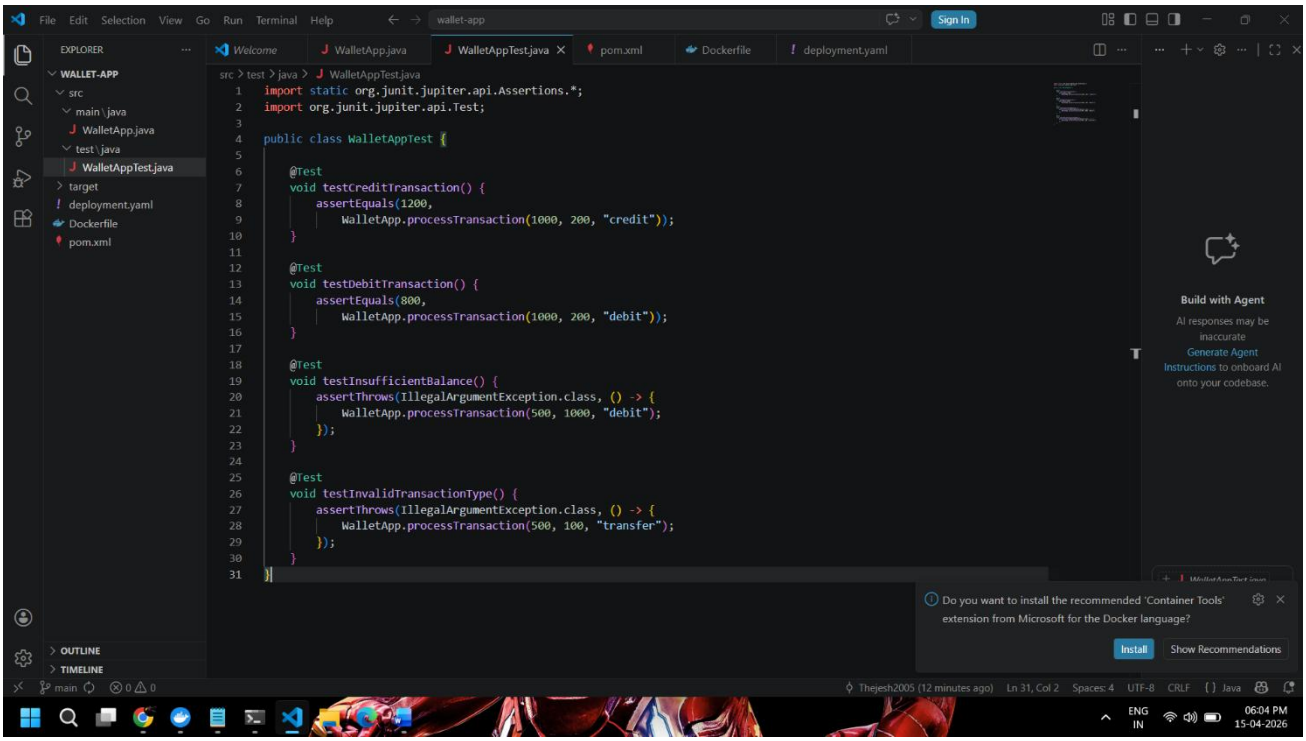


2. Write Junit tests for balance updates

   WalletAppTest.java

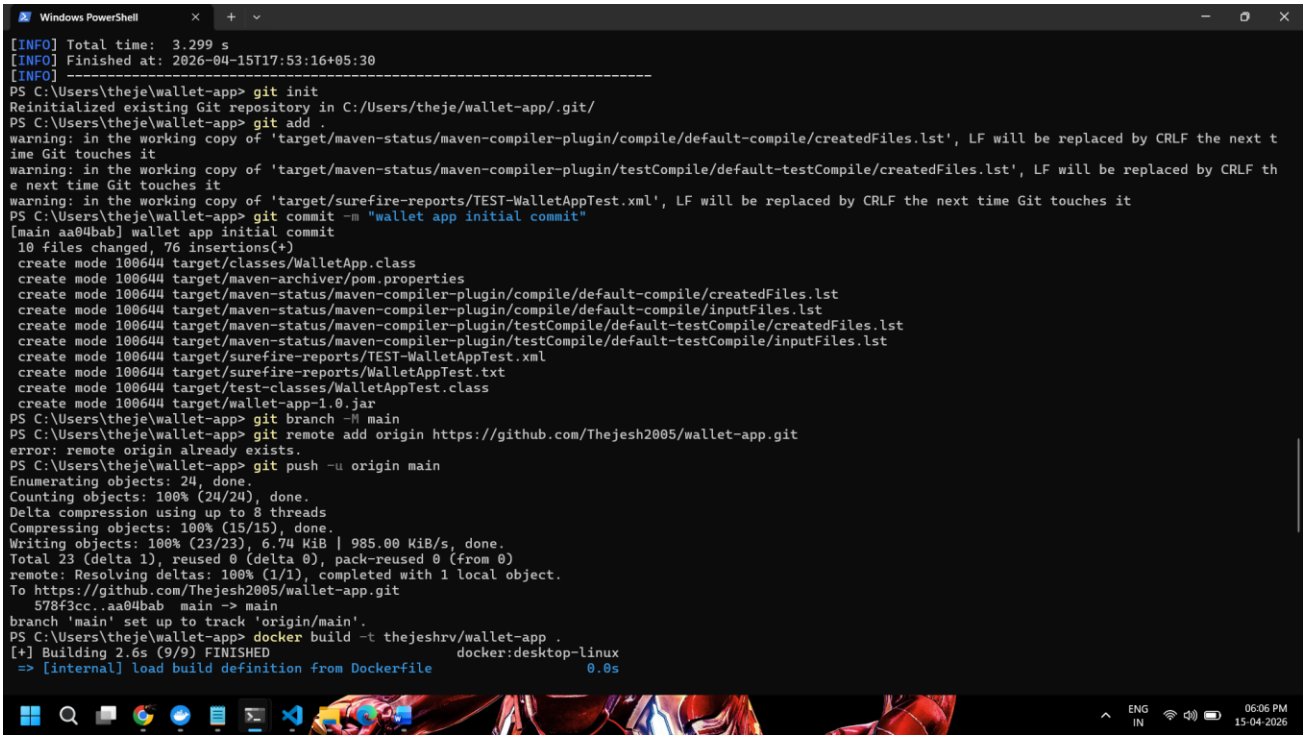
```
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;

public class WalletAppTest {

    @Test
    void testCreditTransaction() {
        assertEquals(1200,
            WalletApp.processTransaction(1000, 200, "credit"));
    }

    @Test
    void testDebitTransaction() {
```

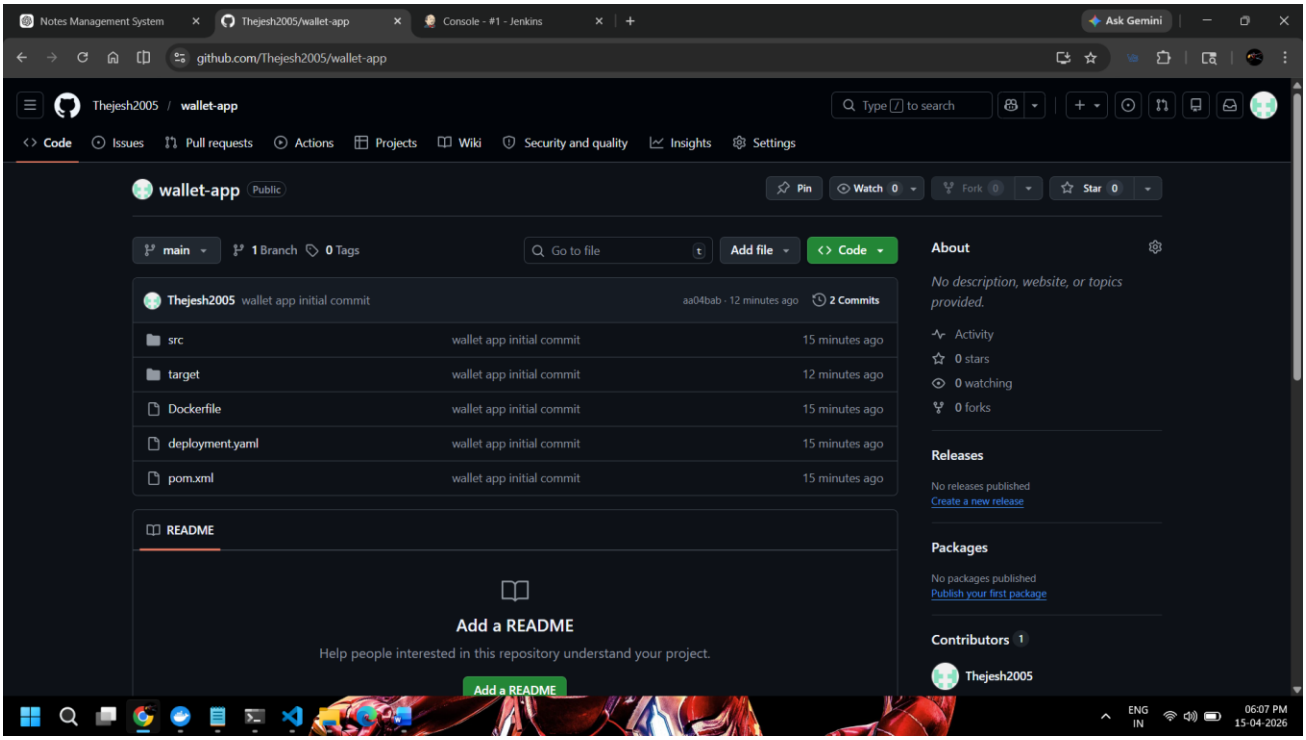
```
        assertEquals(800,
            WalletApp.processTransaction(1000, 200, "debit"));
    }

    @Test
    void testInsufficientBalance() {
        assertThrows(IllegalArgumentException.class, () -> {
            WalletApp.processTransaction(500, 1000, "debit");
        });
    }

    @Test
    void testInvalidTransactionType() {
        assertThrows(IllegalArgumentException.class, () -> {
            WalletApp.processTransaction(500, 100, "transfer");
        });
    }
}
```



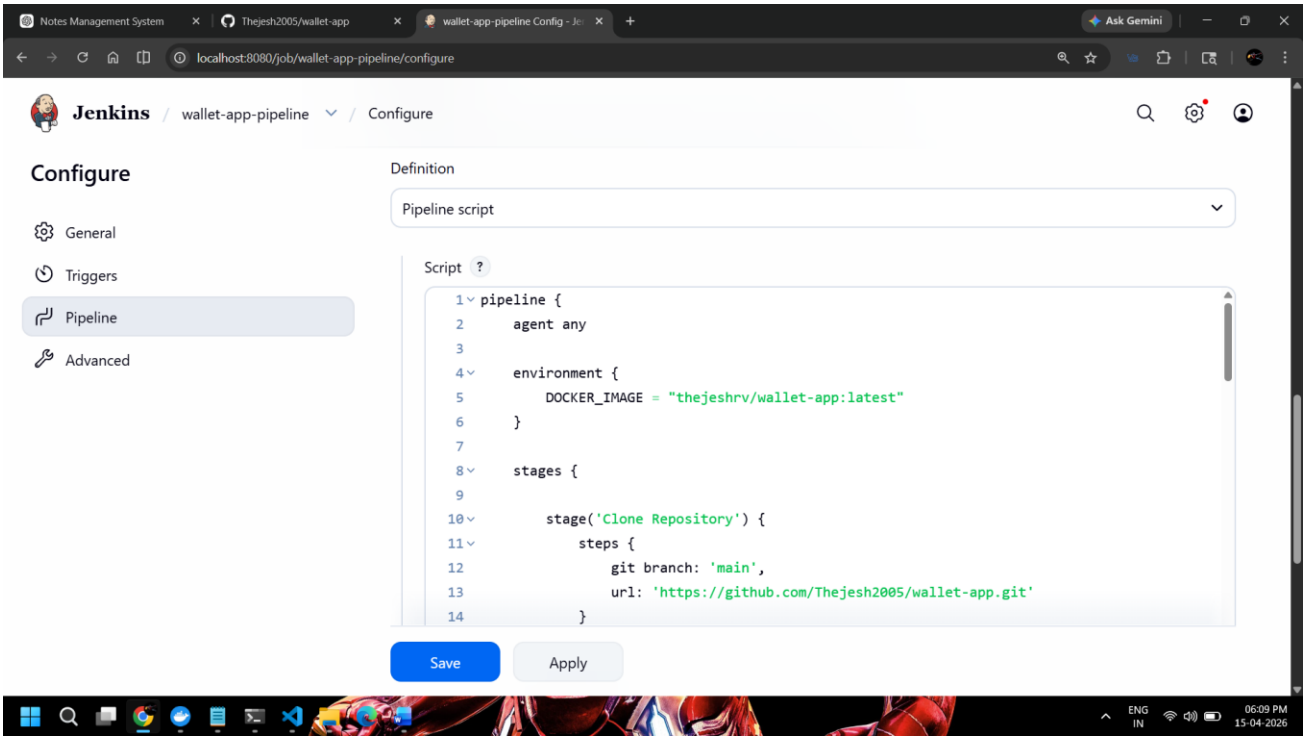
3. Implement CI/CD pipeline with automated testing

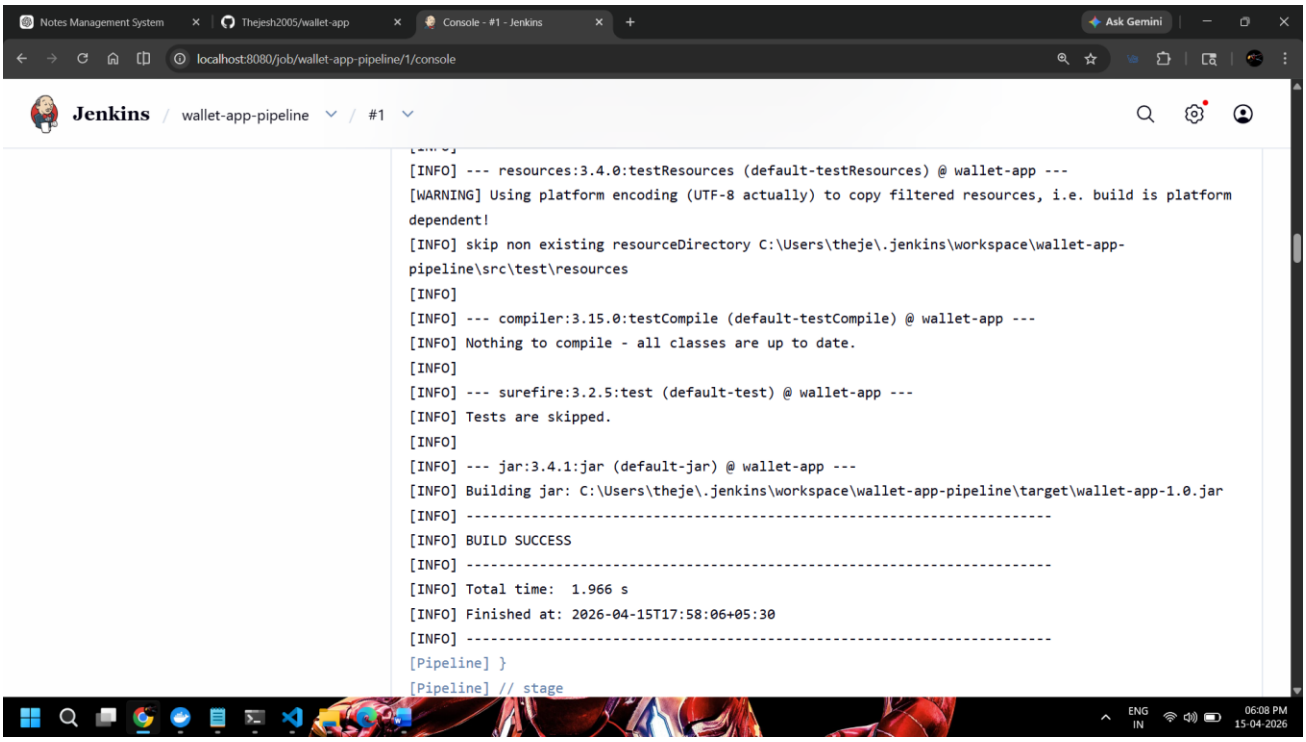
setup version contol using github

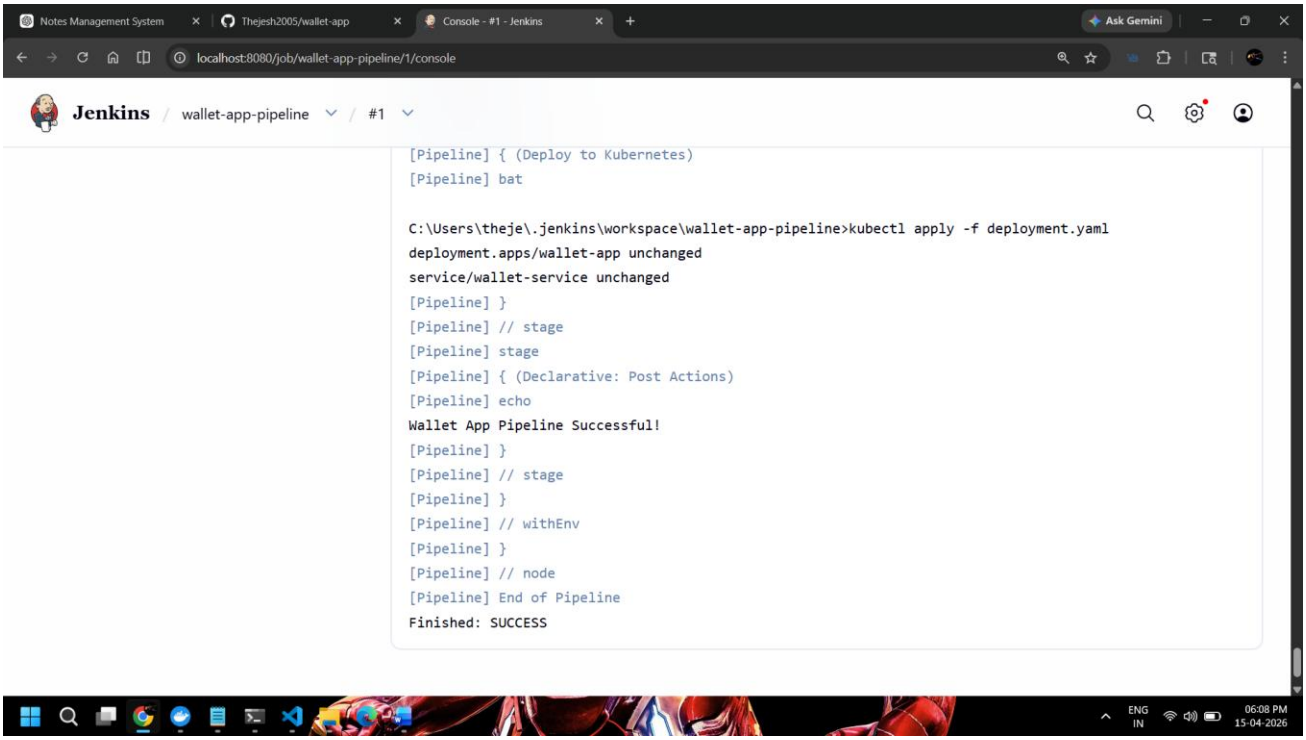




Configure CI pipeline (eg Jenkins /github Actions) to trigger on code Commits , Configure CD pipeline for automatic deployment to staging / production

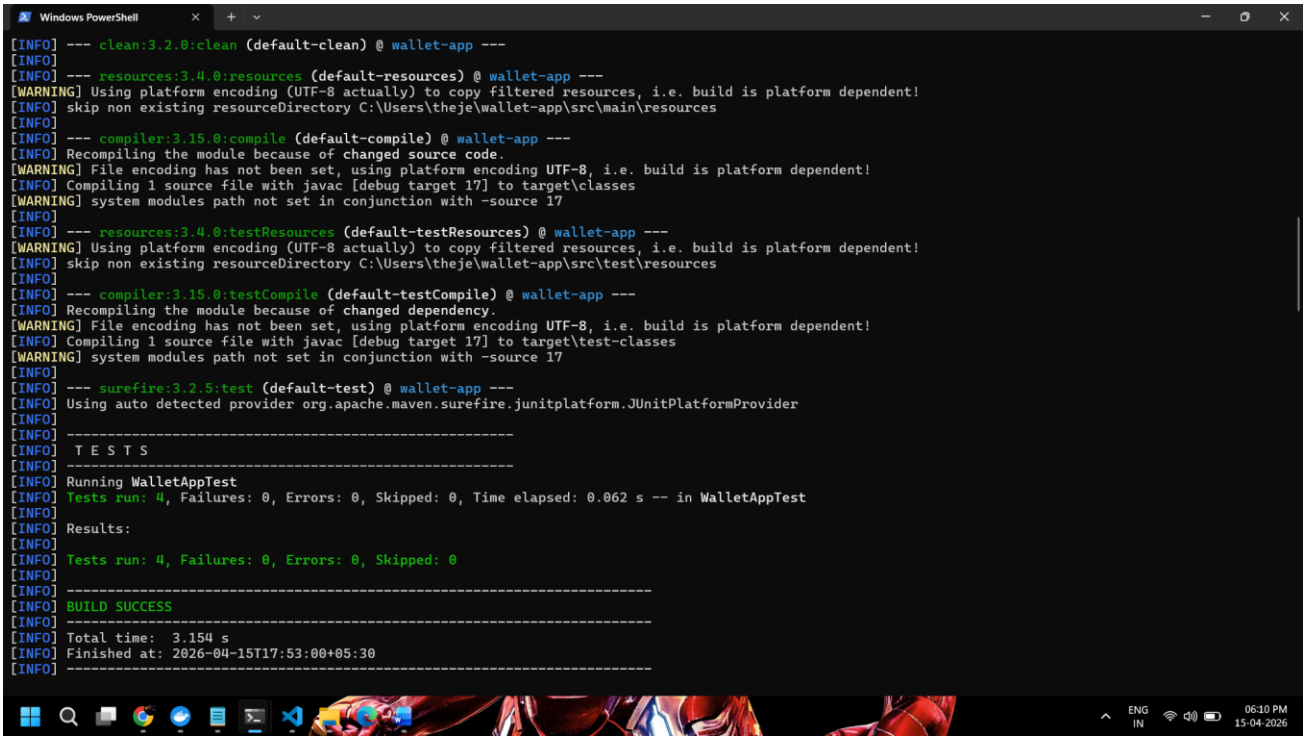

Notes Management System
Thejesh2005/wallet-app
wallet-app-pipeline Config - Je
Ask Gemini
localhost:8080/job/wallet-app-pipeline/configure
Jenkins / wallet-app-pipeline / Configure
Configure
General
Triggers
Pipeline
Advanced
Definition
Pipeline script
Script ?
pipeline {
agent any
environment {
DOCKER_IMAGE = "thejeshrv/wallet-app:latest"
}
stages {
stage('Clone Repository') {
steps {
git branch: 'main',
url: 'https://github.com/Thejesh2005/wallet-app.git'
}
Save
Apply
ENG IN
06:09 PM 15-04-2026


Notes Management System
Thejesh2005/wallet-app
Console - #1 - Jenkins
Ask Gemini
localhost:8080/job/wallet-app-pipeline/1/console
Jenkins / wallet-app-pipeline / #1
[INFO] --- resources:3.4.0:testResources (default-testResources) @ wallet-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\theje\.jenkins\workspace\wallet-app-pipeline\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ wallet-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ wallet-app ---
[INFO] Tests are skipped.
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ wallet-app ---
[INFO] Building jar: C:\Users\theje\.jenkins\workspace\wallet-app-pipeline\target\wallet-app-1.0.jar
[INFO] --
[INFO] BUILD SUCCESS
[INFO] --
[INFO] Total time: 1.966 s
[INFO] Finished at: 2026-04-15T17:58:06+05:30
[INFO] --
[Pipeline] }
[Pipeline] // stage
ENG IN
06:08 PM 15-04-2026



Automate build Process using **maven**

mvn clean test



Mvn clean package

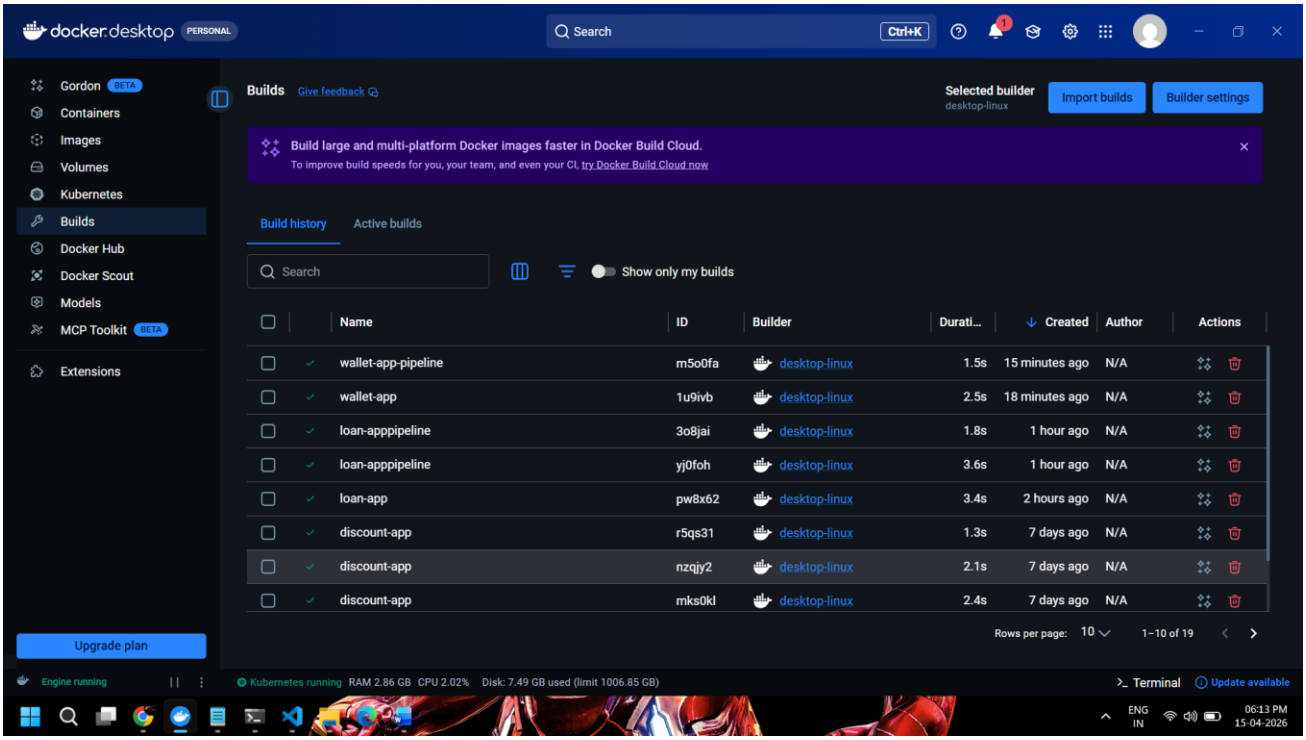
```
[INFO] skip non existing resourceDirectory C:\Users\theje\wallet-app\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ wallet-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ wallet-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\theje\wallet-app\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ wallet-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ wallet-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running WalletAppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.065 s -- in WalletAppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ wallet-app ---
[INFO] Building jar: C:\Users\theje\wallet-app\target\wallet-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.299 s
[INFO] Finished at: 2026-04-15T17:53:16+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\theje\wallet-app> git init
```

Docker

Docker File :

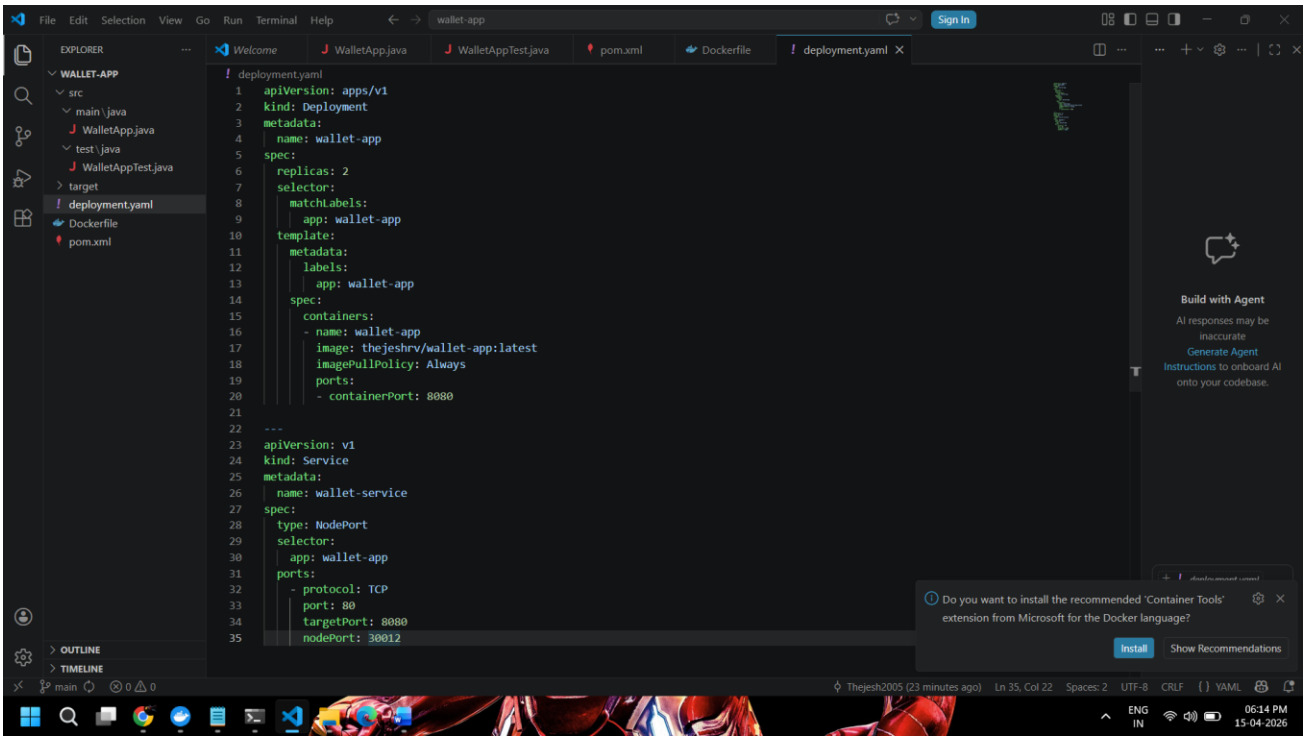
```
FROM eclipse-temurin:17-jdk-alpine

WORKDIR /app

COPY target/*.jar app.jar

EXPOSE 8080

ENTRYPOINT ["java", "-jar", "app.jar"]
```

```
Windows PowerShell
Total 23 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Thejesh2005/wallet-app.git
   578f3cc..aa04bab  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\theje\wallet-app> docker build -t thejeshrv/wallet-app .
[+] Building 2.6s (9/9) FINISHED                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                      0.0s
 => => transferring dockerfile: 175B                                      0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17     2.1s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.i     0.0s
 => [internal] load .dockerignore                                         0.0s
 => => transferring context: 2B                                           0.0s
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk-alpine@sha256     0.0s
 => => resolve docker.io/library/eclipse-temurin:17-jdk-alpine@sha256     0.0s
 => [internal] load build context                                         0.0s
 => => transferring context: 2.64kB                                       0.0s
 => CACHED [2/3] WORKDIR /app                                             0.0s
 => [3/3] COPY target/*.jar app.jar                                       0.0s
 => exporting to image                                                    0.2s
 => => exporting layers                                                   0.1s
 => => exporting manifest sha256:864e41821da3b868aee1bcf92afdbef9ef9c     0.0s
 => => exporting config sha256:d4aa7290b956cd2032a2dab22535e97a1ff968     0.0s
 => => exporting attestation manifest sha256:0e355a485b97ddbecc69e77a     0.0s
 => => exporting manifest list sha256:8f5bd1e5c70ce09b3aec70b661bc90d     0.0s
 => => naming to docker.io/thejeshrv/wallet-app:latest                    0.0s
 => => unpacking to docker.io/thejeshrv/wallet-app:latest                 0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/1u9ivbpjokgpic85yw6s0s6vo
PS C:\Users\theje\wallet-app> docker push thejeshrv/wallet-app
Using default tag: latest
The push refers to repository [docker.io/thejeshrv/wallet-app]
fe04dc30511f: Mounted from thejeshrv/notes-app
589002ba0eae: Mounted from thejeshrv/notes-app
3a547da4fe15: Mounted from thejeshrv/notes-app
7cd2436f004e: Mounted from thejeshrv/notes-app
6e9a775cc420: Pushed
570df2b50426: Pushed
82c540078c28: Mounted from thejeshrv/notes-app
946c51efe0fd: Mounted from thejeshrv/notes-app
latest: digest: sha256:8f5bd1e5c70ce09b3aec70b661bc90da24b4f35532d5f1c1d780a460da7693af size: 856
```

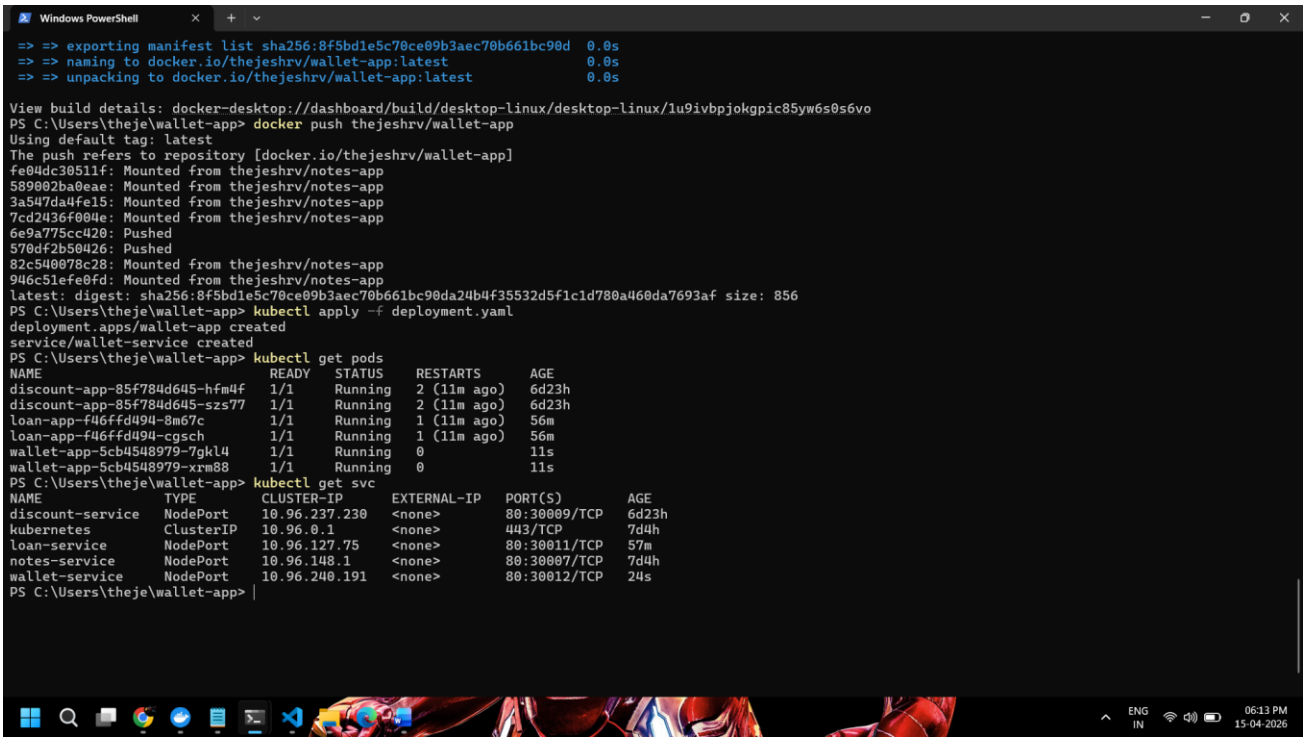


Kubernetes

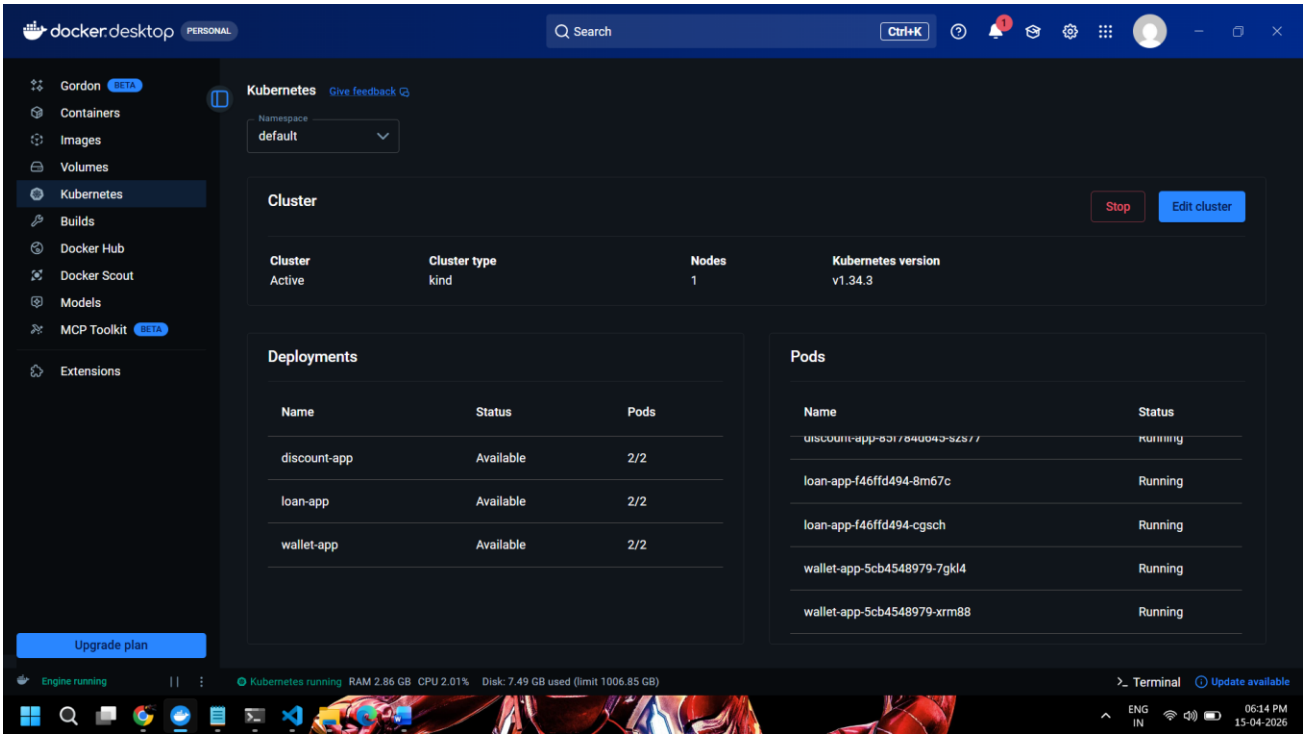
Deployment file :

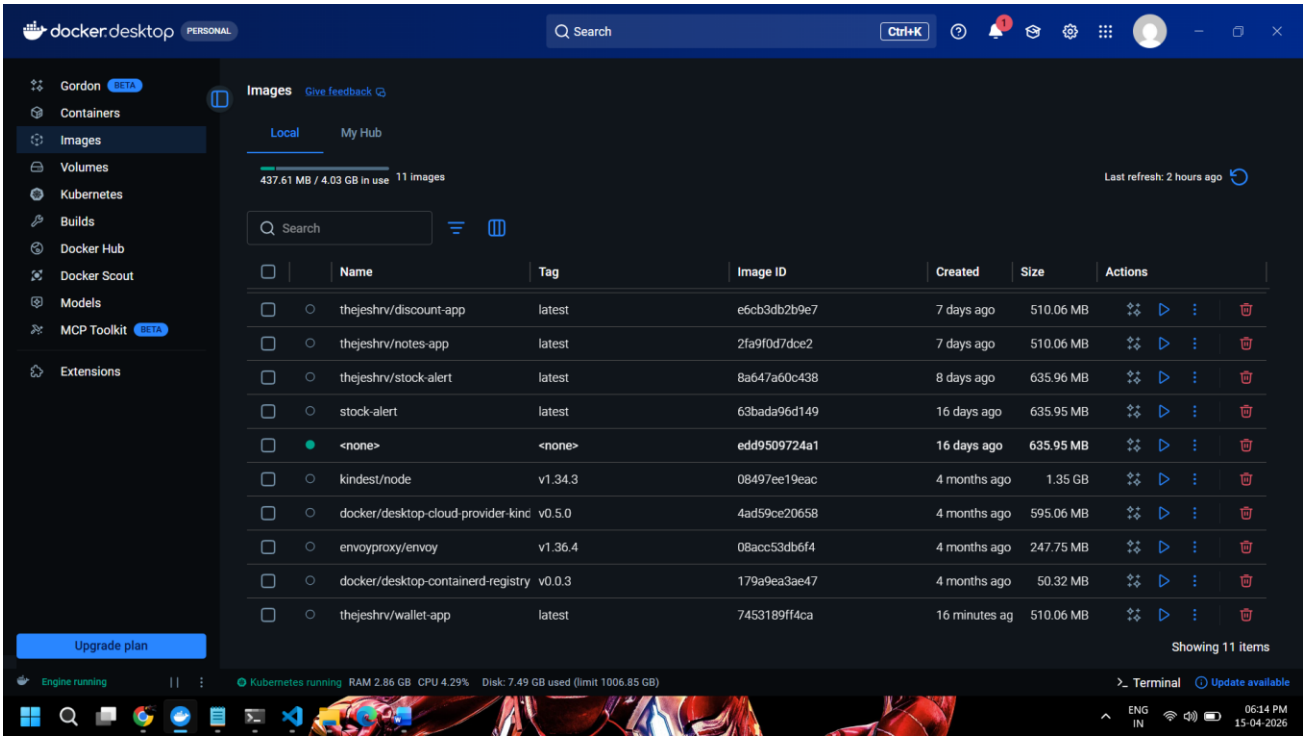


```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: wallet-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: wallet-app
  template:
    metadata:
      labels:
        app: wallet-app
    spec:
      containers:
      - name: wallet-app
        image: thejeshrv/wallet-app:latest
        imagePullPolicy: Always
        ports:
        - containerPort: 8080

---
apiVersion: v1
kind: Service
metadata:
  name: wallet-service
spec:
  type: NodePort
  selector:
    app: wallet-app
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
      nodePort: 30012
```



```
 => => exporting manifest list sha256:8f5bd1e5c70ce09b3aec70b661bc90d  0.0s
 => => naming to docker.io/thejeshrv/wallet-app:latest                 0.0s
 => => unpacking to docker.io/thejeshrv/wallet-app:latest              0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/1u9ivbpjokgpic85yw6s0s6vo
PS C:\Users\theje\wallet-app> docker push thejeshrv/wallet-app
Using default tag: latest
The push refers to repository [docker.io/thejeshrv/wallet-app]
fe04dc30511f: Mounted from thejeshrv/notes-app
589002ba0eae: Mounted from thejeshrv/notes-app
3a547da4fe15: Mounted from thejeshrv/notes-app
7cd2436f004e: Mounted from thejeshrv/notes-app
6e9a775cc420: Pushed
570df2b50426: Pushed
82c540078c28: Mounted from thejeshrv/notes-app
946c51efe0fd: Mounted from thejeshrv/notes-app
latest: digest: sha256:8f5bd1e5c70ce09b3aec70b661bc90da24b4f35532d5f1c1d780a460da7693af size: 856
PS C:\Users\theje\wallet-app> kubectl apply -f deployment.yaml
deployment.apps/wallet-app created
service/wallet-service created
PS C:\Users\theje\wallet-app> kubectl get pods
NAME                          READY   STATUS    RESTARTS      AGE
discount-app-85f784d645-hfm4f   1/1     Running   2 (11m ago)   6d23h
discount-app-85f784d645-szs77   1/1     Running   2 (11m ago)   6d23h
loan-app-f46ffd494-8m67c        1/1     Running   1 (11m ago)   56m
loan-app-f46ffd494-cgsch        1/1     Running   1 (11m ago)   56m
wallet-app-5cb4548979-7gkl4     1/1     Running   0             11s
wallet-app-5cb4548979-xrm88     1/1     Running   0             11s
PS C:\Users\theje\wallet-app> kubectl get svc
NAME              TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
discount-service  NodePort    10.96.237.230   <none>        80:30009/TCP   6d23h
kubernetes        ClusterIP   10.96.0.1       <none>        443/TCP        7d4h
loan-service      NodePort    10.96.127.75    <none>        80:30011/TCP   57m
notes-service     NodePort    10.96.148.1     <none>        80:30007/TCP   7d4h
wallet-service    NodePort    10.96.240.191   <none>        80:30012/TCP   24s
PS C:\Users\theje\wallet-app>
```


docker desktop PERSONAL
Search
Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit BETA
Extensions
Upgrade plan
Kubernetes
Give feedback
Namespace
default
Cluster
Stop
Edit cluster
Cluster
Active
Cluster type
kind
Nodes
1
Kubernetes version
v1.34.3
Deployments
Name
Status
Pods
discount-app
Available
2/2
loan-app
Available
2/2
wallet-app
Available
2/2
Pods
Name
Status
loan-app-f46ffd494-8m67c
Running
loan-app-f46ffd494-cgsch
Running
wallet-app-5cb4548979-7gkl4
Running
wallet-app-5cb4548979-xrm88
Running
Engine running
Kubernetes running RAM 2.86 GB CPU 2.01% Disk: 7.49 GB used (limit 1006.85 GB)
Terminal
Update available
ENG
IN
06:14 PM
15-04-2026


docker desktop PERSONAL
Search
Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit BETA
Extensions
Upgrade plan
Images
Give feedback
Local
My Hub
437.61 MB / 4.03 GB in use
11 images
Last refresh: 2 hours ago
Search
Name
Tag
Image ID
Created
Size
Actions
thejeshrv/discount-app latest e6cb3db2b9e7 7 days ago 510.06 MB
thejeshrv/notes-app latest 2fa9f0d7dce2 7 days ago 510.06 MB
thejeshrv/stock-alert latest 8a647a60c438 8 days ago 635.96 MB
stock-alert latest 63bada96d149 16 days ago 635.95 MB
<none> <none> edd9509724a1 16 days ago 635.95 MB
kindest/node v1.34.3 08497ee19eac 4 months ago 1.35 GB
docker/desktop-cloud-provider-kinc v0.5.0 4ad59ce20658 4 months ago 595.06 MB
envoyproxy/envoy v1.36.4 08acc53db6f4 4 months ago 247.75 MB
docker/desktop-containerd-registry v0.0.3 179a9ea3ae47 4 months ago 50.32 MB
thejeshrv/wallet-app latest 7453189ff4ca 16 minutes ag 510.06 MB
Showing 11 items
Engine running
Kubernetes running RAM 2.86 GB CPU 4.29% Disk: 7.49 GB used (limit 1006.85 GB)
Terminal
Update available
ENG
IN
06:14 PM
15-04-2026